Applied Probability and Simulation

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Chapter 1

Probability

The world around us is uncertain. Just to get a feel of this, let's think of some concrete everyday example. We don't wake up at exactly the same time everyday: even when alarm is on, there would be a difference of a few seconds from day to day. Similarly, the timings of our other activities are often exact, and we sometimes skip some activities like breakfast and exercise (for the author, skipping the latter is the most common). Outside of our own lives, the things like rain and traffic jams are not known *a priori*. The efforts to predict these things have not succeeded to a great extent. Turning our attention to more "scientific" topics, a large number of systems involve some form of randomness. For example, if one decides to study how the smell of a perfume spreads throughout a room, they must accept the randomness involved in the motion of the molecules of the perfume. The same holds for any other gaseous substance and even for molecules in a liquid. Even for solids, the atoms are never at rest, and keep randomly oscillating. The usual notions of 'hot' and 'cold' in fact correspond to the amount of jitters of the atoms such that higher temperatures roughly correspond to higher amount of oscillations.

What I described above should probably make you feel worried: if things are uncertain, how do we ever manage to live our lives at all? And yet, we do seem to be living relatively without much problem in the world filled with randomness! The basic reason that this is possible is that although most particular things are uncertain, their *average* behavior can be predicted with sufficiently high accuracy. To understand this statement properly, we will need to understand how we could talk about uncertainty in a quantitative manner i.e. using numbers. In other words, instead of saying that the thing A is more certain than the thing B, we would like to say that A is 30% more certain than B. Probability theory provides a mathematical language that can be used to quantify uncertainty, and since the whole world involves randomness, we better learn it!

1.1 Random Processes

However, before we dive into the process of quantifying randomness, it would be wise to first think about the concept of randomness itself. All of us are familiar with what randomness means, at least intuitively. In fact, even without studying Probability theory, we do use probability to describe randomness. Let's again consider a few concrete examples:

- When I say "**probably** I will not go to the office tomorrow", I am expressing my inability to predict what I will do tomorrow. This uncertainty is expressed by the use of the word 'probably'. We all, without studying probability theory, can see from this statement that it is more **likely** that I won't go to the office tomorrow than I will go to the office.

- The exact result of a cricket match between Canada and India cannot be predicted before the match, however, given an opportunity to bet on the result, most of us would bet on India. We all do understand, intuitively, that since the Indian cricket team is better than the Canadian team, the **chances** that the Indian team wins are higher.

- If you show a natural number like 109147 to a number theorist and ask her whether it is prime, she might say that **probably** it is but she would need to check using actual division.

In all these examples, when we refer to the probability or chance, what we are really saying is that we are not 100% sure about the outcome. Any process, for which this is true can be **modelled** as a random process or a 'stochastic' process. It could very well be that using some very complicated machinery, sometimes prediction is possible because there is no **intrinsic randomness** in the process. A good example is that of a coin toss, which we will keep using throughout the course. If we know the initial position and the momentum of the coin that we toss, and also the environmental factors like the wind speed, temperature, etc, we should be able to use Newton's laws to exactly predict whether the toss will result into Heads or Tails. However, knowing these things with infinite accuracy is practically impossible, and so is the computation involved. Hence, we can decide to model the coin toss as random although there is no inherent randomness in the system. However, there are situations in which randomness is intrinsic to the system. A good example is the decay of a radioactive atom: it is impossible to predict the time at which it would decay because it is inherently a random system (or that's what the accepted theory of quantum mechanics tells us). We could distinguish these two types of random processes by calling them **pseudo-random** (coin toss) and **random** (decay of an atom). But as long as we decide to model both of these as random processes, it doesn't matter what is the type of the randomness.

1.2 Sample spaces and Events

We will start from two basic definitions which form the basis of the theory of probability: sample space and events.

1.2.1 Sample space

As discussed in the last section, many processes can be modelled as random processes independent of whether they involve real or pseudo randomness. Although the exact output of these processes cannot be predicted, almost always we know the set of all possible outputs for any such process. When you toss a coin, you can't predict whether the output will be Heads or Tails but you know that most probably it is going to be one of these, unless the coin lands on its edge. Similarly, a six-sided die only produces one of the six outputs : $\{1, 2, 3, 4, 5, 6\}$. The set of all possible outputs for a random process is called its **Sample Space**. Usually, a sample space is denoted by Greek letter Ω (pronounced as Omega). Below we give some more examples of random processes and the corresponding sample spaces.

- Tossing a coin: $\Omega = \{H, T\}$
- Throwing a six-sided die: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Measuring the height of a person selected at random from a city: $\Omega = \{x \ : \ x \in (0, \infty)\}$
- Counting the number of friends you have: $\Omega = \{1, 2, 3, \dots\}$
- Selecting a random student from a class and noting down the first letter of their name: $\Omega = \{A, B, C, \dots, X, Y, Z\}$

As you can see from the examples above, a sample space may contain a finite or an infinite number of elements. In fact, we would consider two distinct types of sample spaces.

1.2.2 Countable Sample Spaces

A sample space is said to be countable, if its members can be put in *one-to-one* correspondence with the natural numbers. Here are some examples:

- All sample spaces with a finite number of elements are countable. Thus, sample space for coin tossing $\{H, T\}$ and for a six-sided die $\{1, 2, 3, 4, 5, 6\}$ are countable.
- The sample space $\{1, 2, 3, \dots\}$ that contains all the natural numbers, can obviously be put in one-to-one correspondence with natural numbers, and hence is countable.

- The sample space $\{0, -1, 1, -2, 2, \dots\}$ consisting of all integers is also countable.
- The sample space $\{\frac{a}{b} : a, b \in \mathbb{N}, b \neq 0\}$ consisting of all rational numbers can be put in one-to-one correspondence with natural numbers, and hence is countable.

1.2.3 Uncountable Sample Spaces

A sample space is said to be uncountable, if its members cannot be put in *one-to-one* correspondence with the natural numbers. A necessary (but not sufficient) condition for a sample space to be uncountable is that it should contain infinitely many members. Here are some examples:

- The sample space consisting of all real numbers
- The sample space consisting of all irrational numbers
- The sample space consisting of all positive real numbers

1.2.4 Event

Given a random process and the associated sample space, an event is a subset of the sample space. Let us consider some events for two of the sample spaces described above.

- The sample space for coin toss is $\{H, T\}$. The possible subsets of this are ϕ (an empty set), $\{H\}$, $\{T\}$, and $\{H, T\}$ (the whole set itself). Each of these subsets is an event in this case. Note that H is different $\{H\}$. The former is an element of the sample space while the latter is a set that contains that element. Thus, H is not an event but $\{H\}$ is.
- If we throw a six-sided die and ask whether the outcome is an even number or an odd number, we would basically be asking which of the two events $E = \{2, 4, 6\}$ or $O = \{1, 3, 5\}$ has occurred. Notice how both E and O are subsets of the sample space. We could consider different events in this case. For example, let us ask whether the outcome is greater than 4 or not. The two possible events now are $A = \{1, 2, 3, 4\}$ and $B = \{5, 6\}$. Thus, in case of a given sample space, we can have multiple events.

For a sample space, events that consist of a single element, for example $\{H\}$ or $\{3\}$, are of particular importance. As already explained above, H is different from $\{H\}$, and 3 is different from $\{3\}$. Let us call an event that consists of a single element an **elementary event**.

Given the sample space Ω, consider the set $\mathcal{P}(\Omega)$ of all possible subsets of the sample space. This set is called the **Power set** of Ω. Does every member of $\mathcal{P}(\Omega)$ a valid event? If Ω is countable, it is. However, when Ω is uncountable, some of the subsets of Ω (i.e. some elements of $\mathcal{P}(\Omega)$) cannot be considered as events for reasons that are beyond the scope of this course. This has implications for defining probability for uncountable sample spaces which we will discuss later.

1.2.5 Occurrence of an event

What do we mean when we say "An event A has occurred"? I have seen people getting confused about this innocent looking statement. Worse, when somebody says "Both events A and B have occurred", it leads to extreme confusion for some. It would be best if we understand these statements well before we go to more complex topics.

The statement "An event A has occurred" just means that the output of whatever random process we are considering is in set A. For a six sided die, suppose $A = \{2, 4, 6\}$. Thus, here A is the event that the output is even. Now, if we actually throw a die and find that the number on the top face is 6, we would say that the event A has occurred because A contains 6.

Now consider events $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. The event B can be described as "the output is greater than 3". Suppose we again throw a die and the output is 2, we would say that the event A has occurred but B hasn't since the latter doesn't contain 2. But if the output is 4, then we would say that both events A and

B have occurred since both sets contain 4. Notice that we are talking about a single output but multiple events. When we say that two or more events have occurred, we don't mean that there is more than one outcome!

1.2.6 Combining Events: Union and Intersection

Since events are sets, they can be combined just like any other sets. Two most important ways to combine sets are taking *Union* and taking *Intersection*, defined as follows.

Let $A_1, A_2, \dots$ be sets. Then their *Union*, denoted by $(\bigcup_{i=1}^{\infty} A_i)$ or $A_1 \cup A_2 \cup \dots$, is defined as the set which contains all possible elements from all of these sets:

$$\bigcup_{i=1}^{\infty} A_i = \{x : x \in A_i \text{ for some } i\}$$

Thus, if we take just two sets A_1 and A_2, their union would be the following set:

$$A_1 \cup A_2 = \{x : x \in A_1 \text{ or } x \in A_2 \text{ (or both)}\}$$

Similarly, if $A_1, A_2, \dots$ are sets, their *Intersection*, denoted by $(\bigcap_{i=1}^{\infty} A_i)$ or $A_1 \cap A_2 \cap \dots$, is defined as the set which contains all common elements from all of these sets:

$$\bigcap_{i=1}^{\infty} A_i = \{x : x \in A_i \text{ for all } i\}$$

Thus, if we again take just two sets A_1 and A_2, their intersection would be the following set:

$$A_1 \cap A_2 = \{x : x \in A_1 \text{ and } x \in A_2\}$$

1.2.7 Disjoint Events

Given the events $A_1, A_2, \dots$, we say that they are disjoint, if their intersection is the *Null set* (often denoted by ϕ), which a set that contains no element at all:

$$\bigcap_{i=1}^{\infty} A_i = \phi$$

In other words, if we have two events A and B, and if there are no common elements between them, these two events would be disjoint. This means something very specific: if we know that the event A has occurred, we can completely surely say that the event B has not occurred!

Above, we talked about elementary events, the events that contain only a single element from sample space. Given two elementary events S_1 and S_2, it is easy to see that they are either identical or disjoint! This is because if they are not identical, they must contain two different elements. This is going to be useful to us later while defining probability.

For any set Ω, a **partition** of Ω is a collection of its subsets $Q = \{A_1, A_2, A_3, \dots\}$ that satisfy the following three properties:

1. None of the subsets is empty: $\phi \notin Q$
2. The union of all the sets in Q equals the original set: $\bigcup_i A_i = \Omega$
3. The sets in Q are pairwise disjoint: $A_i \cap A_j = \phi$ whenever $i \neq j$

Later we will need the concept of a partition of a sample space.

1.3 Probability

Now that we are familiar with random/stochastic processes, their sample spaces, and events, we can talk about the core notion of Probability. Let's first define it formally:

Definition 1.3.1. Probability distribution *on a sample space Ω is a function that attaches each event $A \in \Omega$ a real number $\mathbb{P}(A)$, such that the following conditions hold:*

1. $\mathbb{P}(A) \geq 0$
2. $\mathbb{P}(\Omega) = 1$
3. *For any* countable *sequence $A_1, A_2, \ldots$ of disjoint events:*

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A)$$

These axioms are called **Axioms of Probability** or **Kolmogorov's assumptions** in the memory of the great Russian mathematician Andrey Kolmogorov who introduced these in 1933. The definition given above is sufficiently complicated, and it makes sense to try to understand it piece-by-piece. We will do so for countable and uncountable sample spaces separately.

These Axioms immediately imply the following identities:

$$\begin{aligned} \mathbb{P}(A^c) &= 1 - \mathbb{P}(A) \\ \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \end{aligned} \tag{1.1}$$

1.3.1 Probability on countable sample spaces

A countable sample space can be finite or infinite in size (see the examples above). Let us again consider the example of six-sided die whose sample space is finite: $\{1, 2, 3, 4, 5, 6\}$, and consider a function $\mathbb{F}$ that attaches each elementary event $\{1\}, \{2\}, \ldots, \{6\}$ a number $\frac{1}{6}$:

$$\begin{aligned} \mathbb{F}(\{1\}) &= \frac{1}{6} \\ \mathbb{F}(\{2\}) &= \frac{1}{6} \\ \mathbb{F}(\{3\}) &= \frac{1}{6} \\ \mathbb{F}(\{4\}) &= \frac{1}{6} \\ \mathbb{F}(\{5\}) &= \frac{1}{6} \\ \mathbb{F}(\{6\}) &= \frac{1}{6} \end{aligned}$$

To decide whether the function defined above satisfies the assumptions of probability, we will have to first ask whether we would be able to attach such number not just to elementary events, but also to events containing more than one elements such as $\{2, 5\}$ and even to the whole sample space $\{1, 2, 3, 4, 5, 6\}$. A more precise question is if we assume that the third assumption is satisfied by this choice, are first two satisfied? The third assumption gives a recipe to compute the value of the function $\mathbb{F}$ for such events. Recall that elementary events are always disjoint unless they are identical. Hence, we can use the third probability axiom to assign probabilities to events containing more than a single element:

$$\mathbb{F}(\{2, 5\}) = \mathbb{F}(\{2\}) + \mathbb{F}(\{5\}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

It is easy to see that in this case, because every elementary event has been assigned the same value $\frac{1}{6}$, an event containing n elements would be assigned probability $\frac{n}{6}$ which is always non-negative. Hence for the

total sample space Ω, we would have $\mathbb{F}(\Omega) = 1$. In other words, the function $\mathbb{F}$ satisfies all the assumptions of probability, and hence is a valid probability distribution over the sample space $\{1, 2, 3, \ldots, 6\}$.

Now consider a different function on the same sample space:

$$\begin{aligned}
\mathbb{G}(\{1\}) &= \frac{1}{3} \\
\mathbb{G}(\{2\}) &= \frac{1}{6} \\
\mathbb{G}(\{3\}) &= \frac{1}{4} \\
\mathbb{G}(\{4\}) &= \frac{1}{12} \\
\mathbb{G}(\{5\}) &= \frac{1}{12} \\
\mathbb{G}(\{6\}) &= \frac{1}{12}
\end{aligned}$$

If we again use the same technique, we would be able to find $\mathbb{G}(A)$ for any event A (elementary event or not), and it is also easy to verify all the probability assumptions for $\mathbb{G}$. Hence $\mathbb{G}$ also is (another) probability distribution over the sample space $\{1, 2, 3, \ldots, 6\}$. Whenever the sample space is countable and we have a valid probability distribution $\mathbb{P}$ on it, $\mathbb{P}(A)$ for any event A (elementary event or not) is called its **probability**. It should be easy to see now that for a function on a sample space to be a valid probability distribution, the sum of the probabilities for elementary events should be non-zero and should add up to 1 exactly. We can repeat the same excercise for a coin toss:

$$\begin{aligned}
\mathbb{F}(\{H\}) &= 0.7 \\
\mathbb{F}(\{T\}) &= 0.3
\end{aligned}$$

It is easy to check that $\mathbb{F}$ in this case is a valid probability distribution over the sample space $\{H, T\}$. Thus, in this case, the probability of *Heads* is 0.7 and that of *Tails* is 0.3. Incidently, a coin for which the probabilities for Heads and Tails are different is called a **biased** coin whereas a coin for which they are same (both 0.5) is called a **fair** coin.

Let us now consider a sample space that is infinite but countable: $\mathbb{N} = \{1, 2, 3, \ldots\}$ which is the set of all natural numbers. As an example of a valid probability distribution on this sample space, let's consider a function $\mathbb{P}$ as:

$$\mathbb{P}(\{n\}) = \frac{6}{\pi^2 n^2}$$

for every elementary event $\{n\}$. We can again assign values to events consisting of more than one element using the third axiom as we did before. You should take time to verify that this then defines a valid probability distribution on the sample space $\mathbb{N}$. Thus, in this case we can talk about the probability of any event including the elementary events.

Now let us consider a truly complicated sample space: all rational numbers in the open interval $(0, 1)$. Because the set is countable, we should be able to define a valid probability distribution over it. To do that, let us first ask what are the elementary events in this case. We know that a rational number can be written as p/q where both p and q are integers and $q \neq 0$. Since we want only those rational numbers which lie in $(0, 1)$, the choice of p and q cannot be arbitrary. First observe that either p or q cannot be negative although both can be negative. But in that case the number represented would simply be $|p|/|q|$. Hence let us restrict ourselves to the case where both are positive ($p = 0$ case produces 0 which we want to exclude). Also, because the number must be smaller than 1, we must have $p < q$. With these constraints, let us define set A_k which contains all the rational numbers p/q in their lowest terms satisfying these constraints and for which $p + q = k$. Clearly, A_0, A_1 and A_2 are empty because of the constraints we have imposed (verify this!).

A few next sets are:

$$A_3 = \left\{\frac{1}{2}\right\}$$
$$A_4 = \left\{\frac{1}{3}\right\}$$
$$A_5 = \left\{\frac{1}{4}, \frac{2}{3}\right\}$$
$$A_6 = \left\{\frac{1}{5}\right\}$$
$$A_7 = \left\{\frac{1}{6}, \frac{2}{5}, \frac{3}{4}\right\}$$

Of course we have infinitely many of these sets. Thus, we can take elements from these sets in a consecutive fashion and list them like this:

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \frac{1}{5}, \frac{1}{6}, \frac{2}{5}, \frac{3}{4}, \ldots$$

For q_n, the nth number in this list, let us define a function: $\mathbb{P}(\{q_n\}) = \frac{1}{2^n}$. You should invest time in verifying that this function is indeed a valid probability distribution over the set of rational numbers in $(0, 1)$. Thus, even in this case, we can clearly state the probability value for every elementary event, for example:

$$\mathbb{P}\left(\left\{\frac{2}{3}\right\}\right) = \frac{1}{2^4}$$

because $2/3$ is the fourth element in our list.

A note about notation: For an elementary event $\{x\}$, we have been using the notation $\mathbb{P}(\{x\})$ to define its probability. However, most authors simplify the notation by writing this as $\mathbb{P}(x)$. This is fine as long as it doesn't cause any confusion. However, this simplification in the notation doesn't work while writing the probability for an event containing more than one elements from the sample space: we must denote the probability of the event $\{x, y\}$ by $\mathbb{P}(\{x, y\})$, not by $\mathbb{P}(x, y)$ which means a completely different thing as we will see when we will discuss multi-dimensional events.

1.3.2 Probability on uncountable sample spaces

The way the three axioms of the probability work when the sample space is uncountably infinite, is fundamentally different from the countable case. Before going into the technicalities, it would be better to start with an example. Similar to the case of all rational numbers in $(0, 1)$ discussed above, let us consider the sample space made up of all real numbers in $(0, 1)$ (i.e. rationals and irrationals). Can we, similar to the case of the rational numbers' sample space, assign a number $\mathbb{P}(\{x\})$ to each real number x in this sample space and make sure that $\mathbb{P}$ is a valid probability distribution? To answer this, we again need to see how the third axiom can be satisfied. For example, let us consider two events that are clearly disjoint:

$$A = \{x \in \mathbb{R} : x \in (0, 0.5)\}$$
$$B = \{x \in \mathbb{R} : x \in [0.5, 1)\}$$

The event A consists of all the real numbers in the open interval $(0, 0.5)$ (i.e. excluding the endpoints), whereas the event B consists of all the real numbers in the semi-open interval $[0.5, 1)$ which includes 0.5 but excludes 1. Hence:

$$\mathbb{P}(A \cup B) = \mathbb{P}(\{(0, 1)\}) = 1$$

If $\mathbb{P}$ is a valid probability distribution over $(0, 1)$, then the third Kolmogorov assumption now requires that:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) = 1$$

Thus, we must now bear the burden of somehow assigning values to $\mathbb{P}(A)$ and $\mathbb{P}(B)$ such that the above equation is satisfied. In the case of countable sets, we used a trick using which we were able to assign $\mathbb{P}(A)$ for any event A by decomposing the event into elementary events and adding the $\mathbb{P}$ for all the elementary events. This was possible because elementary events are disjoint and whenever we have a countable set, we can always sum its elements. Can we do the same here? In particular, can we add $\mathbb{P}(x)$ for all $x \in (0, 0.5)$ and get the value of $\mathbb{P}(A)$?

The answer is No! Unfortunately, if a set contains *uncountably* many elements, its impossible to sum its elements. This is because to sum the elements of a set, we should be able to pick the first element, followed by the second element, third element, and so on. But the fact that this is impossible for uncountable sets means that we can't add $\mathbb{P}(x)$ for all $x \in (0, 1)$. In fact, once we assign $\mathbb{P}$ to every elementary event $\{x\}$, and declare that $\mathbb{P}(\{(0, 1)\}) = 1$, there is no way we can assign $\mathbb{P}$ to any event that consists of all the points in a subinterval (a, b) of $(0, 1)$ by adding $\mathbb{P}(x)$ for all the x in that subinterval because any such set is uncountable.

The way out of this problem is to attach a different quantity called **probability density**, usually denoted by $p(x)$, to each elementary event $\{x\}$. Notice that this quantity is defined **only** for elementary events, and not for other events. Thus, we can also view it simply as a function of x. Also, $p(x)$ must be non-negative and is assumed to be integrable. Now, given this, for any interval (a, b) of real line, we define the probability corresponding to it as:

$$\mathbb{P}(\{(a, b)\}) = \int_{x=a}^{b} p(x)dx \tag{1.2}$$

In our example, the sample space is the interval $(0, 1)$, and hence by the second Kolmogorov assumption, its probability must be equal to 1. Thus, putting $a = 0$ and $b = 1$ in the definition above, we see that:

$$\mathbb{P}(\{(0, 1)\}) = \int_{x=0}^{1} p(x)dx = \mathbb{P}(\Omega) = 1 \tag{1.3}$$

Thus, $p(x)$ can be any integrable function with only two constraints:

1. $p(x) \geq 0$
2. Its integral over the whole sample space should be equal to 1

Does the definition given in Eq(1.2) satisfy Kolmogorov's Probability axioms? Since $p(x) \geq 0$, its integral is also non-negative, and hence the first axiom is satisfied. We have also constrained $p(x)$ so that its integral over the whole sample space equals 1, so by definition the second axiom is satisfied. To verify that the third axiom is also satisfied, let's first consider the case where the sample space is a finite interval (a, b) of the real line. Now consider any finite partion $\{I_1, I_2, \ldots, I_n\}$ of the sample space (we have defined the meaning of partition before) as defined below:

$$\begin{aligned} I_1 &= (a, x_1), \\ I_2 &= [x_1, x_2], \\ &\vdots \\ I_i &= [x_i, x_{i+1}], \\ &\vdots \\ I_n &= [x_{n-1}, b] \end{aligned} \tag{1.4}$$

Now consider:

$$\sum_{k=1}^{n} \mathbb{P}(I_k) = \int_{a}^{x_1} p(x)dx + \int_{x_1}^{x_2} p(x)dx + \cdots + \int_{x_{n-1}}^{b} p(x)dx = \int_{a}^{b} p(x)dx = 1$$

In particular, if we take a collection of any number of disjoint contiguous intervals $I_1, I_2, \ldots, I_n$, because of the property of integrals that they can be added whenever the upper limit of the previous integrals is

equal to the lower limit of the next integral, we can find out the probability of their union $\mathbb{P}(I_1 \cup I_2 \cup \cdots \cup I_n)$ by adding the individual probabilities. What if the intervals in our collectionare not contiguous? Then as long as the intervals are disjoint, we *define* the probability of their union as the sum of the probabilities for each interval in the collection! This way we can assign probability to any interval on the real line and to the union of several disjoint intervals. This also tells us how we can assign probability to the union of two intervals I_1 and I_2 when they are not disjoint. In this case, we actually can construct a collection of three disjoint intervals: $I_1 \cap I_2$, $I_1 - (I_1 \cap I_2)$, and $I_2 - (I_1 \cap I_2)$. You can easily verify that the union of these three intervals is same as $I_1 \cup I_2$. Thus, we can assign probability to $I_1 \cup I_2$ by adding the probabilities for those three disjoint intervals! This can be easily extended to any number of intervals, overlapping or not.

When probability density is defined on the whole real line or a part of the line (e.g. $(0, 1)$), we can graphically represent $p(x)$ as a function of x. Then the probability of a particular interval (a, b) on the real line is given by the area under the curve $p(x)$ from $x = a$ to $x = b$ because the area is same as the integral in Eq(1.2). Now can we talk about a probability of an elementary event $\{x\}$? This event can be thought of as an interval $[x, x]$ with zero length. The area under the curve is then just zero, and hence $\mathbb{P}(\{x\}) = 0$ for every elementary event! This might be extremely confusing, but there are no incosistencies here. When the sample space is uncountably infinite, non-zero probability can be assigned only to some events like an interval (a, b). In fact, there are certain events in this case which cannot be assigned any probability at all! Such events are called non-measurable, but we won't have to worry about those in this course because the only events we would be dealing with in this course are those that are intervals or their unions.

1.4 Conditional probability, Independence, and Bayes theorem

Imagine that you are playing a simple game with your friend that involves throwing a six-sided die. Your friend throws the die and looks at the outcome, but doesn't reveal it. Instead, she tells you some event that has occurred. For example, she might tell you that the outcome is greater than 3 (event A). Now, she asks you whether the outcome is even (event B). To answer this, you must compute the probability of this second event *given* the first has occurred before you answer her question. This probability is called the **Conditional probability** of event B *given* A, and is denoted by $\mathbb{P}(B|A)$. Notice that there is only one outcome, not two as we have already discussed. In this particular example, we have been told that $A = \{4, 5, 6\}$ has occurred. Since there are two even numbers in this set, $\mathbb{P}(B|A) = 2/3$. We must pay attention to a very important thing here: the probability of event B has changed *after* we knew that A has occurred! If we did not know that A has occurred, then the probability that the outcome is even would have been $3/6 = 1/2$ because there are 3 even outcomes out of 6. The information that A has occurred, thus restricts the sample space and hence potentially changes the probability of all other events. To put this differently, $\mathbb{P}(B)$ is different from $\mathbb{P}(B|A)$.

The conditional probability $\mathbb{P}(B|A)$ can be related to $\mathbb{P}(A \cap B)$ which is the probability that both A and B have happened. Again, make sure that you don't confuse the number of events with the number of outcomes: there is only one outcome. Thus, while computing $\mathbb{P}(B|A)$, we can be sure that event A has occurred, and hence all we need to do, is to find out the fraction of outcomes in A that are also in B. In other words,

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \tag{1.5}$$

1.4.1 Independent events

The Eq(1.5) can be rearranged as:

$$\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A) \tag{1.6}$$

This equation holds for arbitrary events A and B. We also discussed above that the probability of event B in general changes if we know that another event A has occurred. But now consider a special pair of events A and B such that the probability of event B does not change even when we know that A has occurred. That is, $\mathbb{P}(B|A) = \mathbb{P}(B)$. In such case, we say that A and B are **Independent events**. We thus see that,

when A and B are independent,

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A) \tag{1.7}$$

This equation provides a straightforward way to check whether given two events are independent or not: if the probability $\mathbb{P}(A \cap B)$ is equal to the probability $\mathbb{P}(B)\mathbb{P}(A)$, then A and B are independent, otherwise they are not. As an example, consider the following two events for a six-sided die: $A = \{1, 2\}$ and $B = \{2, 3, 4\}$, which means that $\mathbb{P}(A) = 1/3$ and $\mathbb{P}(B) = 1/2$. Thus, $A \cap B = \{2\}$ and hence, $\mathbb{P}(A \cap B) = 1/6 = \mathbb{P}(A)\mathbb{P}(B)$. Hence, A and B are independent. Here we can also see that if somebody tells us that A has happened, then we know that the outcome is either 1 or 2. Therefore, the probability that the outcome is in B is 1/2 as 2 is in B but 1 is not. But we knew that $\mathbb{P}(B) = 1/2$ even if we are not told that A has occurred!

We have earlier discussed the concept of disjoint events. Suppose A and B are disjoint. Can they be independent also? The answer is **No!** The reason is that if A and B are disjoint, and if we know that A has already happened, then we surely know that B has NOT happened. Hence the probability of B changes after we know that A has happened (in fact, its probability becomes zero). This means that A and B cannot be independent.

1.4.2 Bayes theorem

From Eq(1.6), we have:

$$\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

Since there is nothing special about A and B, we can also write this as:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B)$$

Since the LHS of both these is same, equating the Right Hand Sides, and rearranging, we get:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)} \tag{1.8}$$

This innocent looking formula, called **Bayes theorem**, turns out to be so powerful, that it has literally revolutionized many fields and technologies that we use today, and there is special branch of probability and statistics called **Bayesian Inference** that it has given rise to. The important thing that you should pay attention to about this formula is that it allows us to compute the *Inverse Conditional Probability*: if you know $\mathbb{P}(A|B)$ but you are actually interested in $\mathbb{P}(B|A)$, then you can use Bayes formula to compute it.

Example: Box A contains 5 black balls and 5 white balls whereas the box B contains 8 black balls and 2 white balls. You choose one of the boxes uniformly randomly without looking at it and then draw a ball from it. The drawn ball happens to be white. What is the probability that it came from box B?
Solution: It is easy to see that:

$$\mathbb{P}(\text{white}|B) = 0.2$$

. But we want to compute $\mathbb{P}(B|\text{white})$. By Bayes' theorem:

$$\mathbb{P}(B|\text{white}) = \frac{\mathbb{P}(\text{white}|B)\mathbb{P}(B)}{\mathbb{P}(\text{white})} \tag{1.9}$$

To compute the denominator on the RHS, observe that there are total 20 balls out of which 7 are white. If we imagine that all of them were together, then the probability that we would draw a White ball would thus be 7/20. Also, the probability of picking each box is same, and hence $\mathbb{P}(B) = 1/2$. Thus, we have:

$$\mathbb{P}(B|\text{white}) = \frac{\frac{2}{10} \times \frac{1}{2}}{\frac{7}{20}} = \frac{2}{7} \tag{1.10}$$

Is this reasonable? There are less white balls in B, so if we have drawn a white ball, then it is should be more likely that it has come from box A instead of B. Hence, the probability that it has come from B is less than 50%.

Chapter 2

Random Variables

The members of a sample space need not always be numbers, and can also contain information that is not needed for a particular purpose. For example, suppose we are interested in knowing how many heads will appear when we toss five fair coins. The answer could be anything from 0 to 5, and hence there are 6 different possibilities. However, it is easy to see that there are total 32 elements in the sample space, and several different of these outcomes can correspond to the same number of coin tosses: HHT, HTH, and THH all correspond to the occurrence of 2 Heads, and the information of the order in which they occur is not relevant when it comes to knowing this number. Thus, it makes sense to map all the outcomes in this sample space to numbers. Apart from reducing the size of the space of outcomes, this also makes the mathematical handling of the outcomes easier. This mapping from outcomes to numbers is called a **Random variable**.

Definition: A random variable $X : \Omega \to \mathbb{R}$ is a function that assigns each outcome $\omega \in \Omega$ to a real number.

We will follow the convention of using capital letters (X) to denote random variables and small letters (x) to denote a possible value taken by it.

The set of all possible values of a random variable X is called its **Support** and is denoted by $\text{Supp}(X)$. The random variable X is called **discrete** if $\text{Supp}(X)$ is finite or countable, otherwise it is called **continuous**.

A discrete random variable is described by specifying the probability of each outcome, and the function that specifies this probability is called the **Probability Mass Function** or **PMF** for short. We will denote the PMF of a discrete random variable X by f_X. Thus,

$$f_X(x) = \mathbb{P}(X = x) \tag{2.1}$$

A continuous random variable is described by its **probability density function** or **PDF**. We will denote the PDF of a continuous random variable X by p_X. Thus,

$$\int_a^b p_X(x) = \mathbb{P}(a < X \leq b) \tag{2.2}$$

The phrase **Probability distribution** of X is used to mean the PMF when X is discrete and the PDF when X is continuous. Both discrete and continuous random variables can be described using their **Cumulative Distribution Function** or CDF which is defined as:

$$F_X(x) = \mathbb{P}(X \leq x) \tag{2.3}$$

which is the probability that X takes value less than or equal to x. Note that the CDF is defined for all the values from $-\infty$ to ∞, not only for values in $\text{Supp}(X)$. It is monotonically increasing and bounded between 0 and 1.

Ex: Let X be the number of Heads in two tosses of a fair coin. The sample space in this case is:

$$\Omega = \{HH, HT, TH, TT\} \tag{2.4}$$

Thus, $\text{Supp}(X) = \{0, 1, 2\}$ and $f_X(0) = \frac{1}{4}$, $f_X(0) = \frac{1}{2}$, $f_X(0) = \frac{1}{4}$. Also then the CDF of X is given by:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 0.25 & \text{for } 0 \leq x < 1 \\ 0.75 & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases} \tag{2.5}$$

It is also easy to see that if X is a continuous random variable, then

$$p_X(x) = \frac{dF_X}{dx} \tag{2.6}$$

2.1 Important Random Variables

2.1.1 Discrete random variables

1. $X \sim \text{Bernoulli}(p)$ with support $\{0, 1\}$

$$f_X(1) = p \quad \text{and} \quad f_X(0) = 1 - p$$

2. $X \sim \text{Binomial}(n, p)$ with support $\{0, 1, 2, \ldots, n\}$

$$f_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

3. $X \sim \text{Poisson}(\lambda)$ with support $\{0, 1, 2, \ldots\}$

$$f_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

2.1.2 Continuous random variables

1. Uniform random variable: $X \sim \text{U}(a, b)$ with support $(a, b]$

$$p_X(x) = \frac{1}{b-a}$$

2. Exponential random variable: $X \sim \text{Exp}(\lambda)$ with support $(0, \infty)$

$$p_X(x) = \lambda e^{-\lambda x}$$

3. Gaussian/Normal random variable: $X \sim \mathcal{N}(\mu, \sigma^2)$ with support $(-\infty, \infty)$

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

2.2 Joint distribution of X_1 and X_2

Suppose X_1 and X_2 are two discrete random variables. It is quite common for us to want to know not just the probability that X_1 takes some value x_1 or X_2 takes some value x_2, but also the probability that "X_1 takes value x_1 *and* X_2 takes value x_2". For example, if we are playing a game with two six-sided dice, we might be interested in knowing the probability that the combination $(6, 6)$ occurs. The same is true even for continuous random variables, the only difference is that then we are interested in knowing the probability that X_1 takes value in some range $(x_1, x_1 + dx_1)$ *and* X_2 takes value in some other range $(x_2, x_2 + dx_2)$. As an example, if we treat height and weight of a person as continuous random variables, we are often interested in knowing the probability that they have weight and height in certain ranges. This requires us to generalize the notion of a probibility distribution to pairs, and is known as the 'Joint distribution'.

When X_1, X_2 are discrete, the joint distribution is the Joint Probability Mass Function (Joint PMF) $f_{X_1,X_2}(x_1, x_2)$ defined as:

$$f_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2)$$

And when X_1, X_2 are continuous, the joint distribution is specified by the Joint Probability Density Function (Joint PDF) $p_{X_1,X_2}(x_1, x_2)$ so that:

$$\int_{x_1=a_1}^{b_1} \int_{x_2=a_2}^{b_2} p_{X_1,X_2}(x_1, x_2) dx_1 dx_2 = \mathbb{P}(a_1 < X_1 \leq b_1), a_2 < X_2 \leq b_2)$$

When we sum (or integrate) the Joint probability distribution over one of the two random variables, we get the **Marginal distribution** of the other random variable in the pair (X_1, X_2). Thus, for discrete X_1, X_2, the marginal distributions (or just marginals) are given by:

$$f_{X_1}(x_1) = \sum_{x_2} f_{X_1,X_2}(x_1, x_2) \quad \text{and} \quad f_{X_2}(x_2) = \sum_{x_1} f_{X_1,X_2}(x_1, x_2)$$

and for continuous random variables X_1, X_2, marginals are given by:

$$p_{X_1}(x_1) = \int_{x_2} p_{X_1,X_2}(x_1, x_2) dx_2 \quad \text{and} \quad p_{X_2}(x_2) = \int_{x_1} p_{X_1,X_2}(x_1, x_2) dx_1$$

For discrete X_1, X_2, the conditional distribution of X_1 given X_2 is defined as:

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_2}(x_2)} \tag{2.7}$$

And for continuous X_1, X_2, the conditional distribution of X_1 given X_2 is defined as:

$$p_{X_1|X_2}(x_1|x_2) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_2}(x_2)} \tag{2.8}$$

Let's just focus on the discrete case because the continuous case is analogous. Rewriting Eq(2.7), we get:

$$f_{X_1,X_2}(x_1, x_2) = f_{X_1|X_2}(x_1|x_2) f_{X_2}(x_2) \tag{2.9}$$

If the probability of the occurrance of X_1 is not affected by X_2 and vice versa, then we say that they are **independent**. Thus, when X_1 and X_2 are independent,

$$f_{X_1|X_2}(x_1|x_2) = f_{X_1|X_2}(x_1|x_2) = f_{X_1}(x_1)$$

Thus, putting this in Eq(2.9), we see that when X_1 and X_2 are independent,

$$f_{X_1,X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2) \tag{2.10}$$

In other words, when X_1 and X_2 are independent, their joint distribution can be written as the product of their marginals.

2.3 Transformations of Random Variables

Let X be a random variable, and define a new random variable $Y = h(X)$ where h is some function. Since X and Y are not numbers, we must first assign some meaning to this mapping. Suppose we draw samples $x_1, x_2, \ldots, x_n$ from the distribution of X, then we can compute $h(x_1), h(x_2), \ldots, h(x_n)$. Then the random variable Y is defined to be the variable whose samples are these numbers. Here we want to see how to compute the distribution of Y if we know the distribution of X.

To start with, let us assume that X is discrete. Now suppose that there are several values x_i of X such that $h(x_i) = y$. Then, whenever any of these values is sampled from X, the value y gets sampled from Y. Therefore, the probability that y is sampled from Y is equal to the sum of the probabilities of all these x_i. Thus,

$$f_Y(y) = \sum_x f_X(x)\delta(h(x), y) \tag{2.11}$$

where $\delta(a, b) = 1$ if any only if $a = b$, otherwise it is zero.

Example: Let X be a uniform random variable on the set $\{-2, -1, 0, 1, 2\}$. Thus, $f_X(x) = \frac{1}{5}$ for each $x \in \text{Supp}(X)$. Let $Y = X^2$. What is the probability distribution of Y?

Solution: Observe that $\text{Supp}(X) = \{0, 1, 4\}$. We can also see that there are two values each which get mapped to 1 and 4after squaring, while only one value that gets mapped to 0. Hence adding the probabilities, we get:

$$f_Y(0) = \frac{1}{5}, f_Y(1) = \frac{2}{5}, f_Y(4) = \frac{2}{5}$$

Even when X is continuous, conceptually we still have the same problem. However, it is much easier to compute the distribution of Y by first computing its CDF. Thus, we follow the following steps:

1. Write the expression for the cumulative distribution function (CDF) of Y
2. Rewrite the expression using the CDF of X and compute the CDF of X
3. Differentiate the CDF to get the density of Y

Let us see how this works with the help of the following example:

Example: Let $X \sim U(0, 1]$ and $Y = X^2$. Compute the probability density of Y.

Solution: Let us first write the expression for the CDF of Y:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y)$$

To write this in terms of the CDF of X, we observe that since $\text{Supp}(X) = (0, 1]$, the square function is invertible on this support. Hence,

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X \leq \sqrt{y}) = F_X(\sqrt{y})$$

For $x \in (0, 1]$, the CDF of X is given by:

$$F_X(x) = \int_0^x p_X(u)du = \int_0^x 1 \cdot du = [u]_0^x = x$$

Of course, for $F_X(x) = 0$ for $x \leq 0$ and $F_X(x) = 1$ for $x \geq 1$. Hence for $y \in (0, 1]$, we have:

$$F_Y(y) = \sqrt{y}$$

Now we can compute the density of Y by differentiating $F_Y(y)$:

$$p_Y(y) = \frac{dF_Y}{dy} = \frac{1}{2\sqrt{y}}$$

Observe that $p_Y(y) \to \infty$ as $y \to 0$. Thus, even though the probability can never exceed 1, the density function can have arbitrarily large values.

Distribution of the Minimum and the Maximum

Example: Let $X_1, X_2 \sim U(0,1)$ be independent. Let $X_{max} = \max(X_1, X_2)$ and $X_{min} = \min(X_1, X_2)$.
Solution: We have,

$$F_{X_{max}}(x) = \mathbb{P}(X_{max} \leq x) = \mathbb{P}(\max(X_1, X_2) \leq x)$$
$$\therefore F_{X_{max}}(x) = \mathbb{P}(X_1 \leq x, X_2 \leq x)$$

Since X_1 and X_2 are given to be independent, this is same as:

$$F_{X_{max}}(x) = \mathbb{P}(X_1 \leq x)\mathbb{P}(X_2 \leq x) = F_{X_1}(x)F_{X_2}(x)$$

Since both $X_1, X_2 \sim U(0,1)$, for $x \in (0,1]$, the CDFs $F_{X_1}(x) = F_{X_1}(x) = x$. It is easy to see that $F_{X_{max}}(x) = 0$ for $x \leq 0$, and $F_{X_{max}}(x) = 1$ for $x > 1$. Hence for all $x \in (-\infty, \infty)$,

$$F_{X_{max}}(x) = F_{X_1}(x)F_{X_2}(x) = x^2$$

Thus, the density function of X, when $x \in (0,1]$ is:

$$p_{X_{max}}(x) = \frac{dF_{X_{max}}}{dx} = 2x$$

Also, when $x \notin (0,1]$, $F_{X_{max}} = 0$

2.4 Expectation Value and Covariance

Given a random variable X and its CDF $F_X(x)$, we are often interested in the "average value" of X. This is because since X is random, it usually takes a range of values with different probabilities, and although its probability distribution completely describes it, we would like to have some type of summary of the distribution in terms of a single number. The concrete example would be the human height: each person has a different height, but if we were to describe how tall human species is to an alien who has never seen us, then we would say something like "five feet". Expectation value is just this summarization of different values taken by a random variable. Let's start by considering a very simple random variable that takes only two values 1 and 0 (latter we are going to discuss this random variable in more detail). It would be reasonable to say that average of these two values is 0.5. However, this is not entirely right, because computing an average by adding all the values taken by X and then dividing by the total number of values misses an important thign. Assume that X is such that 1 appears most of the time, say 99% of the time, while 0 appears only 1% of the time. Would it still be fine to say that the average value of X is 0.5? Since most of the outcomes are 1, the average should probably be close to 1. Thus, it is important not just to take into account the possible values of X but also their probabilities. Hence, while computing the expectation value, we multiply each possible value x of X by its probability and then add these products. So if X takes value 1 with probability 0.99 and 0 with probability 0.01, then its average value would be: $(0.99 \times 1 + 0.01 \times 0) = 0.99$.

The phrase "Expectation Value" is thus a misnomer as it may mislead us to think that we somehow expect this value from X. In our example, there is no way X would output 0.99 as it can only produce 0 and 1. However, this phrase has got stuck and we have to use it. But always keep in mind that "Expectation Value of X" just means the average value of X, no more no less. Often we also use the term **"Mean value of X"** or just **"Mean of X"** as a synonym for Expectation value and this turns out to be a much better name. We will keep using the phrases "Expectation Value", "Expectation", "Mean Value", and "Average Value" of X interchangably.

Now let us formally define what we have discussed so far about the expectation value. First consider a *discrete* random variable X with PMF f_X. Then the expectation value, denoted by $\mathbb{E}(X)$ is given by:

$$\mathbb{E}(X) = \sum_x x f_X(x) \tag{2.12}$$

where the sum goes over all the values in $\text{Supp}(X)$. This formula can be generalized to the case of a continuous random variable X with probability density p_X as follows. We know that the probability that X takes value in the interval $(x, x + dx]$ is $p(x)dx$. Thus, analogous to above formula, we can just take sum of $xp(x)dx$ over the support of X, and that will give us the expectation value of X. But since X is continuous, instead of sum, we will have an integral. Hence,

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x p_X(x) dx \tag{2.13}$$

The limits of integration are generic in this case. It might be that X takes values only in say an interval $[a, b]$, but then because p_X would be zero outside this interval, the above formula still works. Or in such cases we can shrink limits to an appropriate range.

A note about Notation: The formulas for the expectation values of a discrete and continuous random variables are different: one contains sum while the other contains an integral, one contains PMF, while the other contains PDF. Hence, while talking about the expectation value, it is cumbersome to specify each time whether we are talking about a discrete random varaible or a continuous one. But it is possible to do better: although we need a PMF to describe a discrete random variable while a PDF to describe a continuous random varible, as we have seen, both can be described using the Cumulative Distribution function or the CDF $F_X(x)$ which is in fact defined for all the values in $(-\infty, \infty)$. We can thus use the following notation to denote the expectation value of X independent of whether X is discrete or continuous.

$$\mathbb{E}(X) = \int x dF_X(x) \tag{2.14}$$

It is important to keep in mind that this is a notation, and there is no deeper meaning to it. It just means that if X is discrete, then it is equivalent to Eq(2.12) and if X is continuous, then it is equivalent to Eq(2.13).

It is possible that a random variable X is hybrid, in that it is partly continuous and partly discrete. For example, suppose that with probability 0.8, X takes value in the interval $(-1, 1]$ such that all the values in the interval are equally likely, while with probability 0.1 it takes value -2 and with probability 0.1 it takes value 2. Then such variable must be described using both a PMF and a PDF like this:

$$\begin{aligned} f_X(-2) &= 0.1 \\ f_X(2) &= 0.1 \\ p_X(x) &= 0.4 \end{aligned} \tag{2.15}$$

The good thing about the notation (2.14) we introduced above is that it absorbs even such cases.

2.4.1 Expecation of Transformation, aka Law of unconscious statistician

We have seen how to compute the probability distribution of $Y = h(X)$ when the distribution of X is known. Thus, if we now want to compute the expectation value of Y, then we should use this computed distribution, and hence we would have:

$$\mathbb{E}(Y) = \int y dF_Y(y) \tag{2.16}$$

However, we have also seen that computing the distribution of transformation is somewhat complicated, and hence computing the expectation of a transformation should also be complicated. The amazing thing is that

if you are *only* interested in $\mathbb{E}(Y)$, then you actually don't need to compute the distribution of Y at all! This is because, the distribution of X can be directly used to compute this expectation which is given by:

$$\mathbb{E}(Y) = \int h(x) dF_X(x) \tag{2.17}$$

This amazing theorem saves us all the effort of computing the distribution of Y if all we are interested in is just the expectation of Y. But there is a tendency among many users of this theorem to unconsciously treat this as the *definition* of the expectation of $h(X)$ rather than a consequence of the definition of expectation. For this reason, it is also known as the "Law of unconscious statistician" or sometimes of "Lazy" or "Bored" statistician.

2.4.2 Covariance and Variance

Given two random variables X_1 and X_2 with expectations μ_1 and μ_2 respectively, their *Covariance*, denoted by $\text{Cov}(X_1, X_2)$ is defined as:

$$\text{Cov}(X_1, X_2) = \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)] \tag{2.18}$$

If $X_1 = X_2$, then this quantity is called the *Variance* of X, and is denoted by $\text{Var}(X)$. Thus,

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] \tag{2.19}$$

2.5 Properties of Expectation and Variance

Now we will look at how to compute expectations and variances of random variables that are obtained by combining other random variables.

2.5.1 Expectation of a linear combination

Let $X = aX_1 + bX_2$ where a and b are constants. We are interested in knowing the expectation value of X in terms of the expectation values of X_1 and X_2. Note that we are **not** assuming that X_1 and X_2 are independent, and hence in general it is not possible to factor their joint distribution $f_{X_1,X_2}(x_1, x_2)$ as $f_{X_1}(x_1)f_{X_2}(x_2)$. For now, let us assume that X_1 and X_2 are discrete because when they are continuous, apart from changing sums to integrals and PMFs to PDFs, there is nothing different in the two cases. Thus, we have:

$$\mathbb{E}(X) = \sum_x f_X(x)x$$

To bring in the expectations of X_1 and X_2, we must somehow rewrite this in terms of X_1 and X_2. The way to do this is to run over all possible combinations of all the values of X_1 and X_2 and see what value of X each of these combinations produce. Hence:

$$\mathbb{E}(X) = \sum_{x_1}\sum_{x_2} f_{X_1,X_2}(x_1, x_2)(ax_1 + bx_2)$$

$$\therefore \mathbb{E}(X) = \sum_{x_1}\left[ax_1 \sum_{x_2} f_{X_1,X_2}(x_1, x_2) + b\sum_{x_2} x_2 f_{X_1,X_2}(x_1, x_2)\right]$$

$$\therefore \mathbb{E}(X) = a\sum_{x_1}\left[x_1 \sum_{x_2} f_{X_1,X_2}(x_1, x_2)\right] + b\sum_{x_1}\left[\sum_{x_2} x_2 f_{X_1,X_2}(x_1, x_2)\right]$$

$$\therefore \mathbb{E}(X) = a\sum_{x_1} x_1 f_{X_1}(x_1) + b\sum_{x_2} x_2 \left[\sum_{x_1} f_{X_1,X_2}(x_1, x_2)\right]$$

$$\therefore \mathbb{E}(X) = a\sum_{x_1} x_1 f_{X_1}(x_1) + b\sum_{x_2} x_2 f_{X_2}(x_2)$$

$$\boxed{\therefore \mathbb{E}(X) = a\mathbb{E}(X_1) + b\mathbb{E}(X_2)}$$

This is a remarkable result given that we have not assumed that X_1 and X_2 are independent. Also, then it is easy to see that:

$$\boxed{\therefore \mathbb{E}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i \mathbb{E}(X_i)}$$

We express this result by saying that the Expectation is a linear operation.

2.5.2 Expectation of product

If X_1 and X_2 are independent random variables, then the expectation of their product is the product of their expectation. Note that unlike the case of linear combination, we do need independence here. We will only prove it for the discrete case because as usual the continuous case is easy to work out. Let

$$X = X_1 \cdot X_2$$

where X_1 and X_2 independent discrete random variables. Then,

$$\mathbb{E}(X) = \sum_x x f_X(x)$$

$$\therefore \mathbb{E}(X) = \sum_{x_1}\sum_{x_2} x_1 x_2 f_{X_1,X_2}(x_1, x_2) = \sum_{x_1}\left[x_1 \sum_{x_2} x_2 f_{X_1,X_2}(x_1, x_2)\right]$$

Now we can see that unlike the case of linear combination, there is no further simplification possible unless we assume that the joint PMF can be factorized. Hence, let us see what happens if X_1 and X_2 are independent. In that case, $f_{X_1,X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$, and so the above expression simplies as:

$$\mathbb{E}(X) = \sum_{x_1}\left[x_1 \sum_{x_2} x_2 f_{X_1}(x_1) f_{X_2}(x_2)\right] = \sum_{x_1}\left[x_1 f_{X_1}(x_1)\left(\sum_{x_2} x_2 f_{X_2}(x_2)\right)\right]$$

$$\therefore \mathbb{E}(X) = \sum_{x_1}\left[x_1 f_{X_1}(x_1)\mathbb{E}(X_2)\right] = \sum_{x_1}\left[x_1 f_{X_1}(x_1)\right]\mathbb{E}(X_2)$$

$$\boxed{\therefore \mathbb{E}(X) = \mathbb{E}(X_1)\mathbb{E}(X_2)}$$

2.5.3 Variance of a linear combination

Expression for the covariance (and hence for the variance) can also be written in a different form as follows:

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)] = \mathbb{E}[X_1 X_2 - \mu_1 X_2 - \mu_2 X_1 + \mu_1\mu_2]$$

Using the linearity of expectation, we can then write this as:

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mu_1\mathbb{E}(X_2) - \mu_2\mathbb{E}(X_1) + \mu_1\mu_2$$

$$\therefore \mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mu_1\mu_2 - \mu_2\mu_1 + \mu_1\mu_2$$

$$\boxed{\therefore \mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mathbb{E}(X_1)\mathbb{E}(X_2)}$$

Setting $X_1 = X_2 = X$, we thus get:

$$\boxed{\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2}$$

Now let $X = aX_1 + bX_2$ and consider,

$$\mathbb{E}(X^2) = \mathbb{E}[(aX_1 + bX_2)^2] = \mathbb{E}[a^2 X_1^2 + 2abX_1X_2 + b^2 X_2^2] = a^2\mathbb{E}(X_1^2) + 2ab\mathbb{E}(X_1X_2) + b^2\mathbb{E}(X_2^2)$$

and

$$[\mathbb{E}(X)]^2 = (a\mathbb{E}(X_1) + b\mathbb{E}(X_2))^2 = a^2[\mathbb{E}(X_1)]^2 + 2ab\mathbb{E}(X_1)\mathbb{E}(X_2) + b^2[\mathbb{E}(X_2)]^2$$

Subtracting these two expressions, we get:

$$\mathrm{Var}(X) = a^2\left[\mathbb{E}(X_1^2)\right]^2 + b^2\left[\mathbb{E}(X_1^2)\right]^2$$

It is straightforward to show that, in the general case of linear combination of n random variables:

$$\boxed{\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i,j} a_i a_j \mathrm{Cov}(X_i, X_j)} \tag{2.20}$$

The importance of the covariance of two random variables is that it measures the amount of linear dependence between them. Assume that the random variables X_1 and X_2 are such that $X_2 = aX_1 + b + Y$ where a and b are constants. Here Y is a Gaussian random variable that is independent of X_1, has zero mean, and has variance σ^2. Thus,

$$\mathbb{E}(X_1X_2) = \mathbb{E}(aX_1^2 + bX_1 + X_1Y_1) = a\mathbb{E}(X_1^2) + b\mathbb{E}(X_1) + \mathbb{E}(X_1)\mathbb{E}(Y)$$

Here we used the fact that X_1 and Y are independent, and hence $\mathbb{E}(X_1Y) = \mathbb{E}(X_1)\mathbb{E}(Y)$. Moreover, because we have assumed that $\mathbb{E}(Y) = 0$, the last term vanishes. Hence,

$$\mathbb{E}(X_1X_2) = a\mathbb{E}(X_1^2) + b\mathbb{E}(X_1)$$

Also,

$$\mathbb{E}(X_1)\mathbb{E}(X_2) = \mathbb{E}(X_1)(a\mathbb{E}(X_1) + b + \mathbb{E}(Y)) = a\left(\mathbb{E}(X_1)\right)^2 + b\mathbb{E}(X_1)$$

Thus, the covariance of X_1 and X_2 is:

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1X_2) - \mathbb{E}(X_1)\mathbb{E}(X_2) = a\left[\mathbb{E}(X_1^2) - \left(\mathbb{E}(X_1)\right)^2\right] = a\mathrm{Var}(X_1) \tag{2.21}$$

$$a = \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)}$$

Thus, covariance in this case is proportional to the proportionality constant a, which is a coefficient of the linear term X_1. Hence covariance measures the linear dependence between two random variables. In general, a relation between two random variables can be much more complicated with higher order dependencies, but covariance does not measure it.

2.6 Convergence of Random variables

Just like we talk about an infinite sequence of numbers $x_1, x_2, \ldots, x_n, \ldots$, we can talk about a sequence of random variables $X_1, X_2, \ldots, X_n, \ldots$. However, the conceptual meaning of considering such sequence must be made precise before we can work with it, and generalize notions such as *convergence*. Since a random variable doesn't have a fixed value, a sequence of random variables cannot be a particular sequence of numbers. To ascribe a meaning to such sequence we have to think in terms of *sampling* from each of the random variables in the sequence. Imagine that we sample a number from each of the random varialbes $X_1, X_2, \ldots, X_n, \ldots$, and thus we get a sample of numbers $x_1, x_2, \ldots, x_n, \ldots$. If we repeat this process, we will potentially get a different sequence $y_1, y_2, \ldots, y_n, \ldots$. This way, we can form infinitely many sequences of numbers from a given sequence of random varialbes. However, each of these sequences in our collection doesn't have the same probability of showing up because each random variable in our sequence has certain distribution which determines the probability of drawing a given number from it, and each sampled sequence of numbers will have different probability of appearing. Hence the right way to think about a sequence of random variables is to think in terms of a collection of infinitely many sequences of numbers with some probability distribution over this collection.

Henceforth we will assume that the random variables in the sequence $X_1, X_2, \ldots, X_n, \ldots$ are Independent which means that the outcome of sampling from any one of them does not affect the probability of outcome from any other. However, we are not assuming that they are identically distributed (for defining *convergence*, there will be not much use of considering such sequence if they are identically distributed).

To understand the idea of convergence, first let us consider the notion of convergence for a sequence of numbers $x_1, x_2, \ldots, x_n$. When we say that this sequence converges to the number L as $n \to \infty$, what we mean is that the absolute difference $|x_n - L|$ can be made smaller than any $\epsilon > 0$ by choosing n sufficiently large. It doesn't matter how small ϵ is, as long as it is greater than 0. Thus, as we choose larger and larger values of n, the values of x_n will become closer and closer to L, and we write:

$$\lim_{x \to \infty} x_n = L$$

As we discussed above, the sequence $X_1, X_2, \ldots, X_n$ is really a collection of many sequences of numbers, even if one of the sequences actually converges to some number L, there is no guarantee that other sequences in the collection will do the same. Hence, there doesn't exist more than one notions of convergence for a sequence of random variables. We will only talk about two of these.

2.6.1 Convergence in distribution

We say that the sequence of random variables $X_1, X_2, \ldots, X_n$ converges *in distribution* to the random variable X, written as $X_n \rightsquigarrow X$, if for all $x \in (-\infty, \infty)$,

$$\lim_{n \to \infty} F_{X_n}(x) = F_X(x)$$

Here $F_X(x)$ is the CDF of X. This means that we choose larger and larger values of n, the shape of the probability distribution of X_n looks more and more similar to the probability distribution of the random variable X. For example, suppose

$$X_n \sim \mathcal{N}\left(\mu + \frac{1}{n}, \sigma^2 + \frac{1}{n}\right)$$

and let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then it is easy to see that X_n converges to X *in distribution* as $n \to \infty$. Notice however that this **does not** mean that if we sample two numbers from X_n and X, then for sufficiently large n, these numbers will be close to each other. This is because even if two random variables X and Y have the same probability distribution, sampling from them will in general output different numbers.

2.6.2 Convergence in probability

We say that the sequence of random variables $X_1, X_2, \ldots, X_n$ converges *in probability* to the random variable X, written as $X_n \xrightarrow{p} X$, if for any $\epsilon > 0$, however small, we have:

$$\lim_{n \to \infty} \mathbb{P}(|X_n - X| < \epsilon) = 1$$

This equation also needs to be understood in terms of sampling. The expression $|X_n - X|$ is the absolute difference of the random variables X_n and X. Thus, to compute its value, imagine that we sample number x_n from X_n and number x from X, and then take the absolute difference $|x_n - x|$. If we repeat this, process, this absolute difference probably will have a different value. Thus, $|X_n - X|$ is a random variable. Hence for a fixed $\epsilon > 0$, we can talk about the probability that this random variable has value less than ϵ. We say that X_n converges to X in probability if the limit of $\mathbb{P}(|X_n - X| < \epsilon)$ converges to 1 as $n \to \infty$. Thus, in this case the values taken by X_n and those taken by X tend to get closer and closer with high probability. For example, if $X_n = X + \frac{1}{n}$, then the difference between the values taken by X and X_n is $\frac{1}{n}$. Hence in the limit of large n, this difference goes to zero, and hence $X_n \xrightarrow{p} X$.

The convergence in probability is a stronger form of convergence than the convergence in distribution because if $X_n \xrightarrow{p} X$ then $X_n \rightsquigarrow X$, but the opposite need not be true.